General Research & Structured Data Engine

Complete implementation plan for a source-controlled research system that turns natural-language requirements into verified, structured outputs such as Google Sheets.

What is being built

A general-purpose AI-assisted research engine - not limited to universities, schools, or education. A user can ask for almost any structured research task, optionally provide files and source URLs, define strict source rules, and receive a validated Google Sheet with field-level sources and review parameters.

Examples of supported research

Core principle

The system should optimize for trustworthy, source-compliant data - not for filling every cell at any cost. If a requested field is restricted to a source and that source does not contain the information, the correct output is N/A.

Primary outcome

A repeatable workflow in which the user gives requirements once; the engine plans and performs the research, records provenance, validates results, generates the final Google Sheet, and allows the user to delete temporary research data after approval.

1. User Flow

The user completes the full workflow from a single Streamlit interface. The research target can come from an uploaded file, a natural-language request, or both.

1	**Input**	Upload CSV/XLSX/PDF or describe the entities to research.
2	**Requirements**	Describe fields, filters, quantities, and desired output.
3	**Source rules**	Add URLs and say which fields must come from which sources.
4	**Plan**	System converts the request into a structured research plan.
5	**Research**	Primary-source research first; missing fields follow allowed fallback rules.
6	**Validate**	Normalize values, verify sources, detect conflicts, assign status.
7	**Export**	Create and format Google Sheet with final data and audit/source tabs.
8	**Approve**	User reviews; click Done & Perfect to delete temporary job data.

Example input

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“Find extracurricular activities for Grades 7-12. I need activity name, category, eligibility, location, online/offline, fee, organizer, website, registration link and contact. Use official organizer sources for eligibility, fee and registration. If a strict-source field is unavailable, put N/A.”
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What the engine derives

Entity	Extracurricular activity
Scope	Grades 7-12; user-defined geography
Fields	Name, category, eligibility, location, delivery mode, fee, organizer, website, registration, contact
Source policy	Official organizer for strict fields; web search where fallback is allowed
Missing behavior	N/A when a permitted source does not provide the field

2. Technical Architecture

The core architecture is domain-agnostic. “University”, “course”, “company”, “activity”, or any other entity is simply a research type generated from the user’s request.

Streamlit UI	**Request Analyzer**	**Research Planner**	**Tavily + URL Research**
Extraction + Normalization	**Validation + Conflict Engine**	**Temporary Job Storage**	**Google Sheets Exporter**

Logical components

Request Analyzer	Reads natural language, uploaded files, field requirements, filters, output expectations, and source constraints.
Schema Builder	Converts the request into an entity type plus a structured list of fields, data types, and validation rules.
Research Planner	Groups research by entity/source rather than performing one independent search per field.
Source Manager	Tracks explicit URLs, allowed domains, preferred sources, strict sources, and fallback permissions.
Extraction Layer	Turns retrieved web content into structured field values plus evidence and source metadata.
Validation Layer	Checks types, source compliance, entity matching, freshness, contradictions, completeness, and missing-data behavior.
Output Layer	Creates a formatted Google Sheet with final data, source/audit information, and review-required records.

3. Research Pipeline & Tavily Strategy

The engine should research at the entity/source level, then selectively fall back only for unresolved fields. This improves consistency and avoids wasteful field-by-field searching.

A. Interpret	Identify entities, fields, constraints, filters, source rules, and requested quantity.
B. Resolve source policy	For every field, assign primary source, allowed fallback, and missing-data rule.
C. Primary-source pass	For each entity, research the relevant pages from the preferred official/source domain and extract all requested fields that are available there.
D. Gap analysis	Mark fields as FOUND, N/A-BY-RULE, or MISSING-NEEDS-FALLBACK.
E. Fallback pass	Only missing fields with fallback permission are researched through broader web search.
F. Validation pass	Validate the extracted values against data type, entity, source, and consistency rules.
G. Finalize	Attach source URLs/evidence to values, calculate status, and prepare the export.

Grouped research rule
For an entity such as a university or activity provider, the engine should discover and retrieve the relevant source pages as a small research set (for example: admissions, fees, eligibility, contact, registration) and extract all requested fields found there. It should not automatically issue one web search per field.

Tavily role

• Basic search: lower-cost discovery when broad coverage is sufficient.

• Advanced search: deeper retrieval when basic search does not provide an adequate result or when the user requests higher research depth.

• URL / page extraction: retrieve content from user-specified or discovered pages so the model can work from source text rather than relying only on search snippets.

• Caching and batching: reuse source results within a job and group retrieval work to reduce duplicated requests.

4. Source Rules, Accuracy & Validation

Accuracy comes primarily from deterministic research rules around the AI, not from asking a language model to “be accurate.”

Strict source	QS Ranking -> QS only	Accept only values supported by the permitted source. Otherwise return N/A.
Preferred + fallback	Tuition -> official site, then web	Exhaust preferred source first; search fallback only for unresolved fields.
Web allowed	City -> web search allowed	Use search and validate the selected source.
User URL	Specific program page attached by user	Use the provided URL/domain as an authoritative source according to the rule.

Validation checks

Check	Purpose
Type / range	Acceptance rate 0-100; rankings numeric; fees currency-aware; emails syntactically valid; URLs valid.
Entity match	Confirm the value belongs to the intended organization/program/location, especially where names are similar.
Source compliance	Ensure a field obtained from a strict source is actually supported by that permitted source.
Conflict detection	When sources disagree, retain candidate values and flag the field instead of silently choosing a number.
Freshness / timestamp	Record when the value was researched so the output is traceable and refreshable.
Missing data	N/A is a valid result when permitted sources do not contain the requested information.

5. Output, Sources & Research Parameters

The Google Sheet is the durable output. The working research job can be temporary and cleaned up after the user approves the result.

Recommended Google Sheet structure

Tab	Contents
Final Data	One row per entity; requested fields in a clean, analysis-ready structure.
Sources / Audit	Entity, field, value, source URL, source type, evidence, retrieved timestamp, confidence/status.
Review Required	Conflicts, low-confidence values, unresolved entity matches, or policy exceptions requiring human review.

Per-field provenance

Every important value should be traceable to its source rather than having one generic “sources” list for the whole row. Example: QS rank -> QS source; tuition -> official institution source; city -> whichever permitted source supplied it.

UI results area

Main result	Google Sheet link, progress, completion state, summary counts.
Sources panel	Relevant source URLs grouped by entity and/or field.
Parameters panel	Fields requested/found, source compliance, high-confidence rate, N/A count, review count, pages analyzed, searches/credits used, timestamps.
Review action	Done & Perfect -> confirm export and delete temporary research data.

Temporary storage strategy
Use temporary job state for active research only. Add an expiration time as a safety net. When the user confirms “Done & Perfect,” delete intermediate research data, cached source content, and job artifacts while leaving the Google Sheet as the durable output.

6. Technology Stack

Python	Core application logic, orchestration, schemas, extraction, normalization, validation, cleanup and integrations.
Streamlit	Local/private web interface for upload, chat-style requirements, URLs, search settings, progress, results and review.
Tavily	Web search and web content retrieval for discovery, deeper search and source extraction.
MongoDB Atlas	Recommended job/state store for the full system; keep active research jobs temporary and use automatic expiration. It can be omitted if simple filesystem state is sufficient, but it is useful for reliable job management.
Google Sheets API	Create the final spreadsheet, write structured results, add multiple tabs, format cells and return/share the sheet.
Pandas / OpenPyXL / PDF tools	Parse CSV/XLSX/PDF inputs and transform them into the internal research dataset.
LLM provider (configurable)	Interpret requests, build schemas, generate targeted queries, extract structured information and explain conflicts. Keep provider-independent rather than hard-coding one model.

Implementation principles

• Domain-agnostic core: no university-specific assumptions in the research engine.

• Source-first research: primary/strict sources are exhausted before permitted fallback.

• Batch/group research: retrieve a useful source set per entity, then extract multiple requested fields together.

• Deterministic validation around AI: LLM output is treated as a candidate result, not as unquestioned truth.

• Provider-independent integrations: keep search and LLM providers replaceable through adapters/configuration.

• Private-by-default operation: run locally during development and minimize retention of source content and intermediate data.

7. Implementation Plan

The build should be organized around the final workflow rather than a narrow first release. The objective is a dependable personal automation tool that can research 50 or 500 entities without manual prompting.

1. Core data model	Define ResearchJob, Entity, FieldRule, SourceRule, Evidence, Result and ValidationStatus models.
2. Request interpreter	Parse natural-language requirements into entities, fields, filters, quantity, source policies and output requirements.
3. Input layer	Support CSV/XLSX/PDF and direct text instructions; normalize entity lists.
4. Research planner	Construct grouped research passes per entity/domain and decide primary vs fallback work.
5. Tavily adapter	Implement search/extract calls, retries, batching, caching within a job, and failure handling.
6. Extraction + normalization	Convert web content into typed field candidates plus evidence and source URLs.
7. Validation engine	Apply field rules, source restrictions, entity matching, conflicts, freshness and N/A behavior.
8. Job/state management	Store active jobs, progress, results and expiry metadata in MongoDB; clean up on approval or expiry.
9. Google Sheets export	Create final, audit and review tabs; format and share the output.
10. Streamlit UI	Build the single-page workflow with request input, URLs, search mode, live progress, sources, parameters and approval actions.
11. Reliability testing	Test missing data, duplicate names, contradictory sources, inaccessible pages, rate limits, partial failures and large batches.

Operational example

User enters a research request for 500 entities -> system generates a plan -> researches each entity from allowed primary sources -> fills gaps only where fallback is permitted -> validates -> writes the Google Sheet -> exposes sources and parameters -> user approves -> temporary job is deleted. The same workflow can be reused for universities, extracurricular activities, courses, products, companies, opportunities, or other structured research tasks.

Success criterion
The system should remove the repetitive human work from research while preserving source discipline. A good run is not “every field is filled”; it is “every returned field is traceable, policy-compliant, validated, and explicit about what could not be verified.”

8. Conclusion

The final system is best understood as a general-purpose research and data-generation engine with strict source controls. Its differentiator is not simply that it can search the web; it is that it can translate a human research request into explicit source policies, group retrieval work intelligently, validate extracted values, preserve provenance, and export the result in a useful structured format.

Recommended target stack

Python + Streamlit + Tavily + MongoDB Atlas + Google Sheets API + a configurable LLM provider, with temporary job retention and automatic cleanup.

Final product definition
“A general-purpose AI-assisted research engine that accepts natural-language requirements, files and optional source URLs; intelligently searches permitted sources; extracts and validates structured information; tracks provenance and uncertainty; and produces a ready-to-use Google Sheet.”

Example domains

Universities • Courses after Grade 10 • Extracurricular activities • Competitions • Scholarships • Profile-building opportunities • Companies • Products • Market research • Directories • Any other task where the user wants structured, source-backed research.

Prepared as the implementation blueprint for the personal research automation system.